bmm testlabs

ENAC ENSAYOS Nº 696/LE1541

Evaluation Report for:
NGT Interactive LLC

Random Number Generator
Vertex version 1.0.1

Manufacturer:	**NGT Interactive LLC**
RNG Name:	**Vertex 1.0.1**
ATF Report Number:	**RNG.GLI.NGTI.1001.01.01**
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BMM Spain Testlabs s.l.u

EVALUATION REPORT

Client name & Address:	NGT Interactive LLC 3111 N. University Dr., Suite 105 33065 - Coral Springs Florida (USA)
Client Reference Number:	Client Submission Letter Dated 18th January 2019
Testing dates:	Start date: 18th January 2019 End date: 30th January 2019
Product / Game Description:	Vertex 1.0.1 RNG SIGNATURES: See paragraph 5
Test Category:	RNG evaluation.
Jurisdictions Recommended:	GLI-19 v 2.0
Technical Standard used for Evaluation:	GLI-19 v 2.0
Location where test was performed:	BMM Spain Testlabs, s.l.u. Parc Tecnològic del Vallés (C.E.N.T.) Avda. Parc Tecnològic del Vallés, 3 08290 – Cerdanyola del Vallés Barcelona – España
Location where report was issued:	BMM Spain Testlabs, s.l.u. Parc Tecnològic del Vallés (C.E.N.T.) Avda. Parc Tecnològic del Vallés, 3 08290 – Cerdanyola del Vallés Barcelona – España
Conclusion:	PASS
BMM Reference Number:	NGTI.1001
Method/Procedures used:	EURAF-SPA-MO-41
Consultant(s):	Slava Kolmykov

1 PURPOSE OF EVALUATION

NGT Interactive LLC has requested BMM to evaluate the random number generator (RNG) Vertex against the GLI.19.

2 DESCRIPTION OF RNG

The RNG is a direct implementation of dev/urandom. Generated numbers are stored in server and then called from there.

3 BMM EVALUATION PERFORMED

BMM examined the RNG source code and performed statistical tests on the output from the RNG. The source code file(s) used are listed in the section 5.

3.1 Source Code Review

The following sections describe the implementation of the RNG in the source code.

3.1.1 SEEDING

The RNG is self-seeding, no seeding algorithm is needed.

3.1.2 CYCLING

The RNG doesn't cycle, and the sequences are not reproducible.

3.1.3 SCALING

The scaling method does not introduce bias.

3.1.4 SECURITY

The RNG is cryptographically secure.

3.2 Statistical Testing

Statistical tests were performed on the output from the RNG. Raw output from the RNG was subjected to a range of tests in the Empirical, Diehard and NIST test suites. Appendix B describes the tests run in each test suite.

4 TEST RESULTS

Each test tests the hypothesis that the RNG is a random source of numbers. A "p-value" is produced for each test run, which is the probability that a truly random process would produce the same or a more extreme result. P-values are expected to be uniformly distributed between 0 and 1. Each test is performed at least 100 times, and the p-values for each test are evaluated using an Anderson-Darling test. This produces a single p-value, which is the probability that the individual p-values have been produced from a uniform distribution.

Finally, the p-values from each test in the same test suite are combined using the Holm-Bonferroni method to provide an overall p-value. This process adjusts each p-value to ensure that the overall probability of accepting the RNG as random matches the confidence interval used. The overall p-value, equal to the minimum of the adjusted p-values, is compared to a specific alpha value to determine if the RNG is accepted or rejected as being random for a specific confidence interval. For a 95% confidence interval, the alpha value used is 0.05.

4.1 Empirical Tests:

- Range [0,31]:

Test	P-values	95% Confidence	99% Confidence
Frequency Test	1,000000	PASS	PASS
Serial Correlation Test	0,351256	PASS	PASS
Runs Test	1,000000	PASS	PASS
Gap Test	1,000000	PASS	PASS
Coupon Collector Test	1,000000	PASS	PASS
Subsequences Test	1,000000	PASS	PASS
Poker Test	0,460992	PASS	PASS
Overall	**0,351256**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

- Range [0,51]:

Test	P-values	95% Confidence	99% Confidence
Frequency Test	1,000000	PASS	PASS
Serial Correlation Test	1,000000	PASS	PASS
Runs Test	1,000000	PASS	PASS
Gap Test	1,000000	PASS	PASS
Coupon Collector Test	1,000000	PASS	PASS
Subsequences Test	1,000000	PASS	PASS
Poker Test	1,000000	PASS	PASS
Overall	**1,000000**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

- Range [0,110]:

Test	P-values	95% Confidence	99% Confidence
Frequency Test	1,000000	PASS	PASS
Serial Correlation Test	1,000000	PASS	PASS
Runs Test	0,845364	PASS	PASS
Gap Test	0,998488	PASS	PASS
Coupon Collector Test	1,000000	PASS	PASS
Subsequences Test	0,408501	PASS	PASS
Poker Test	1,000000	PASS	PASS
Overall	**0,408501**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

- Range [0,999]:

Test	P-values	95% Confidence	99% Confidence
Frequency Test	1,000000	PASS	PASS
Serial Correlation Test	0,869736	PASS	PASS
Runs Test	1,000000	PASS	PASS
Gap Test	1,000000	PASS	PASS
Coupon Collector Test	0,075137	PASS	PASS
Subsequences Test	1,000000	PASS	PASS
Poker Test	1,000000	PASS	PASS
Overall	**0,075137**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

4.2 Diehard Tests

Test	P-values	95% Confidence	99% Confidence
Binary Rank 32x32 Test	1,000000	PASS	PASS
Binary Rank 6x8 Test	1,000000	PASS	PASS
Birthday Spacings Test	1,000000	PASS	PASS
Bitstream Test	1,000000	PASS	PASS
Count The 1's Stream Test	0,952905	PASS	PASS
Count The 1's Specific Test	0,952905	PASS	PASS
Runs Test	1,000000	PASS	PASS
Squeeze Test	1,000000	PASS	PASS
Overall	**0,952905**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

4.3 NIST Tests

Test	P-values	95% Confidence	99% Confidence
Approximate Entropy Test	1,000000	PASS	PASS
Block Frequency Test	1,000000	PASS	PASS
Cumulative Sums Test	1,000000	PASS	PASS
Discrete Fourier Transform Test	1,000000	PASS	PASS
Frequency Test	0,851265	PASS	PASS
Linear Complexity Test	0,539233	PASS	PASS
Longest Run of Ones Test	1,000000	PASS	PASS
Non-Overlapping Template Matchings Test	1,000000	PASS	PASS
Overlapping Template Matchings Test	1,000000	PASS	PASS
Random Excursions Test	1,000000	PASS	PASS
Random Excursions Variant Test	1,000000	PASS	PASS
Rank Test	1,000000	PASS	PASS
Runs Test	1,000000	PASS	PASS
Serial Test	1,000000	PASS	PASS
Universal Test	1,000000	PASS	PASS
Overall	**0,539233**	**PASS**	**PASS**

Conclusion: The RNG is **ACCEPTED** as random at the 95% confidence interval.
Conclusion: The RNG is **ACCEPTED** as random at the 99% confidence interval.

5 SOURCE CODE FILES

The following file(s) are used by the RNG. The signatures provided are generated using HMAC-SHA1 with an empty (0) key.

Files	HMAC-SHA1
vertexrng.exe	84563C134B1BA84ADF299A30C94CBF7F1472C442
CivetServer.cpp	601AC9E0D89C49EBD41B95910CE2541427C467ED
CivetServer.h	F644E181D56867412A79A4432D35D4BCD531B53B
civetweb.c	9DD0E1DEA6AB0C535A85922F4B4F064E55BA50AA
civetweb.h	D387A50DD5B1F1A565E07A600AC49046CA6EACC9
cJSON.c	81EEB93594C5783D0A683962905011860BBC43D2
cJSON.h	3F498618338B32437C7537CE84DD17291DA9EFC9
main.cpp	47AF61F21122EB0CC6A9568293C16EA3DC0B469E

6 ADDITIONAL INFORMATION/OBSERVATIONS

Test Samples were generated directly with manufacturer's code in Linux environment:

- Ubuntu 16.04
- Cmake 3.10

7 CONCLUSION

Accordingly from the test results[1] obtained from the testing performed and results obtained, BMM Spain Testlabs s.l confirms that the item submitted under test conforms to all the relevant GLI.19 requirements described in the Scope section.

Yours faithfully,

MARIO ZILEVSKI

Firmado digitalmente por MARIO ZILEVSKI
Fecha: 2019.02.01 10:48:03 +01'00'

Director of Technical Services - Europe
Mario Zilevski

[1] The results included in this document are referred exclusively to the sampled tested, such as it is described in the corresponding section.

ANNEX: TEST DESCRIPTIONS

The following tests were used to test the statistical properties of the RNG:

- Empirical Tests

  The Empirical Tests are based on the tests described by Donald Knuth in The Art of Computer Programming Volume 2: Seminumerical Algorithms (1997). They test sequences of numbers scaled to specific ranges.

Frequency Test	Counts of each number occurring across the sample set.
Serial Correlation Test	Counts of non-overlapping groups of numbers occurring together. Group sizes of two, three, and four are tested separately.
Runs Test	Counts of ascending and descending sequences of numbers. Note that this is a different test to the Runs Test in the Diehard and NIST Tests.
Gap Test	Counts of the size of gaps between successive occurrences of a given number. Each number in the range is tested separately.
Coupon Collector Test	Counts of sequence lengths required to complete a full set of each number in the range.
Subsequences Test	Similar to the Serial Correlation Test for pairs of numbers, except looking at numbers separated by a specific gap. Step sizes of 5, 10, 15, and 20 are tested separately.
Poker Test	The sequence is split into groups of five. The number of unique values in each group is counted.

- Diehard Tests

  The Diehard Tests are based on the test suite published by George Marsaglia in 1995. They test sequences of raw binary output from the RNG.

Binary Rank 32x32 Test	Matrices are created using 32 32-bit words. The ranks of the resulting matrices are counted.
Binary Rank 6x8 Test	Same as the Binary Rank 32x32 Test, except each matrix is formed using 6 values, each taking 8 bits from successive 32-bit words with a specific offset. All possible offsets are tested separately.
Birthday Spacings Test	26-bit values are taken from successive 32-bit words with a specific offset. The values are sorted, and the spacings between them calculated. The number of spacings of the same size are counted. All possible offsets are tested separately.
Bitstream Test	Blocks of 2^18 values are treated as a stream of overlapping 20-bit values. The number of possible 20-bit values that are not found in each block is counted.
Count The 1's Stream Test	8-bit values are taken and assigned a “letter” based on the number of one’s appearing in the binary representation of each value. Overlapping groups of 5 “letters” are counted.
Count The 1's Specific Test	Similar to the Count the 1’s Stream Test, except 8-bit values are taken from successive 32-bit words with a specific offset. All possible offsets are tested separately.
Runs Test	Counts sequences of increasing and decreasing 32-bit words. Note that this is a different test to the Runs Test in the Empirical and NIST Tests.
Squeeze Test	A value of 2^31 is repeatedly multiplied by 32-bit words, dividing by 232 and taking the ceiling of the result each time. The number of successive words that are required to reduce the value down to 1 is counted, and the value is reset to 231 and the process repeated.

- NIST Tests

  The NIST Tests are based on the suite of tests released by the National Institute of Standards and Technology in Special Publication 800-22, Revision 1a (revised April 2010). They test sequences of raw binary output from the RNG.

Approximate Entropy Test	Similar to the Serial Test, count each possible m-bit values, except it does so for two adjacent m bit lengths and compares the two.
Block Frequency Test	Similar to the Frequency Test, except the data is split into equal size blocks. The number of ones and zeroes in each block is counted.
Cumulative Sums Test	Random walks are created by converting the data to +1 / -1 for 1 / 0 respectively and summing consecutive values.
Discrete Fourier Transform Test	The data is transformed using a Discrete Fourier Transform. The number of peaks within the 95% threshold are counted.
Frequency Test	The number of ones and zeroes in the binary output is counted.
Linear Complexity Test	The length of the linear complexity of the random sequence is determined.
Longest Run of Ones Test	The data is split into equal size blocks. The longest run of ones in each block is determined and counted.
Non-Overlapping Template Matchings Test	The data is split into equal size blocks. Each block is searched for a specific pattern of bits and counted. A separate test is run for various bit patterns. Each bit pattern searched does not overlap with itself. That is, when the pattern is matched, the end of the pattern cannot be the start of another match.
Overlapping Template Matchings Test	Similar to the Non-Overlapping Templates Test, except only one pattern is searched, which may overlap with itself.
Random Excursions Test	As with the Cumulative Sum Test, random walks are created by converting the data to +1 / -1 for 1 / 0 respectively and summing consecutive values. The number of times a given state is visited between returns to zero are counted. Separate tests are run for various states from -4 to +4, not including 0.
Random Excursions Variant Test	Similar to the Random Excursions Test, except the number of times the given state is visited is counted for the entire sequence. Separate tests are run for various states from -9 to +9, not including 0.
Rank Test	Matrices are created using 32 32-bit words. The ranks of the resulting matrices are counted. Note that this is fundamentally the same test as the Binary Rank 32x32 Test in the Diehard Tests, although the implementation may differ.
Runs Test	Runs of consecutive bits of the same value of various lengths are counted.
Serial Test	Counts of each possible m-bit values. Separate tests are run for various m bit lengths.
Universal Test	Distances between repeated patterns of bits are counted.